Value-at-Risk (VaR) Forecast

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1 Data Exploration

1.1 Introduction to this Database

The dataset contains 4,640 daily SPY observations from January 4, 2000 to June 27, 2018, including log returns and two realized volatility measures . The objective is to forecast three lower-tail quantiles of the next trading day's return using information available at the forecast date:

$$q_{\alpha,t+1} = Q_\alpha(r_{t+1} \mid \mathcal{F}_t), \qquad \alpha \in \{0.01, 0.05, 0.10\}.$$

Here, the next trading day's log return is denoted by r_{t+1} and the information set available through the current day is denoted by $\mathcal{F}_t$.

This section examines 1,748 observations from January 4, 2000 to December 29, 2006. The later period is reserved for subsequent rolling out-of-sample evaluation. The analysis focuses on the **return distribution, temporal dependence in volatility, and potential predictive information in the realized measures.**

1.2 Return Distribution

The mean daily return is approximately -0.0015%, which is small relative to the daily standard deviation of approximately 1.1343%. Return skewness is 0.0417, indicating a broadly symmetric distribution over this period. Excess kurtosis is 2.9695, above the normal distribution benchmark of zero, suggesting heavy tails.

The histogram and normal QQ plot support this observation. The central observations lie relatively close to the reference line, while the lower and upper tails depart downward and upward, respectively.

These findings motivate considering Student-t innovations in GARCH and directly estimating the required quantiles.

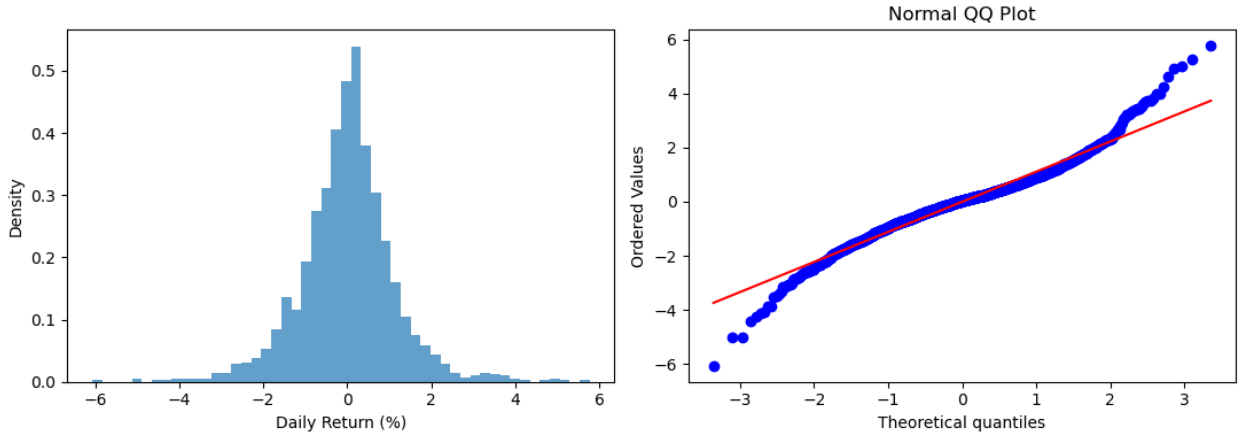


Figure 1: Distribution and QQ plot for returns

1.3 Time-Varying Volatility and Dependence

The return series and trailing 63-day standard deviation show substantial differences in risk levels across periods. VaR forecasts should therefore adjust to recent volatility conditions.

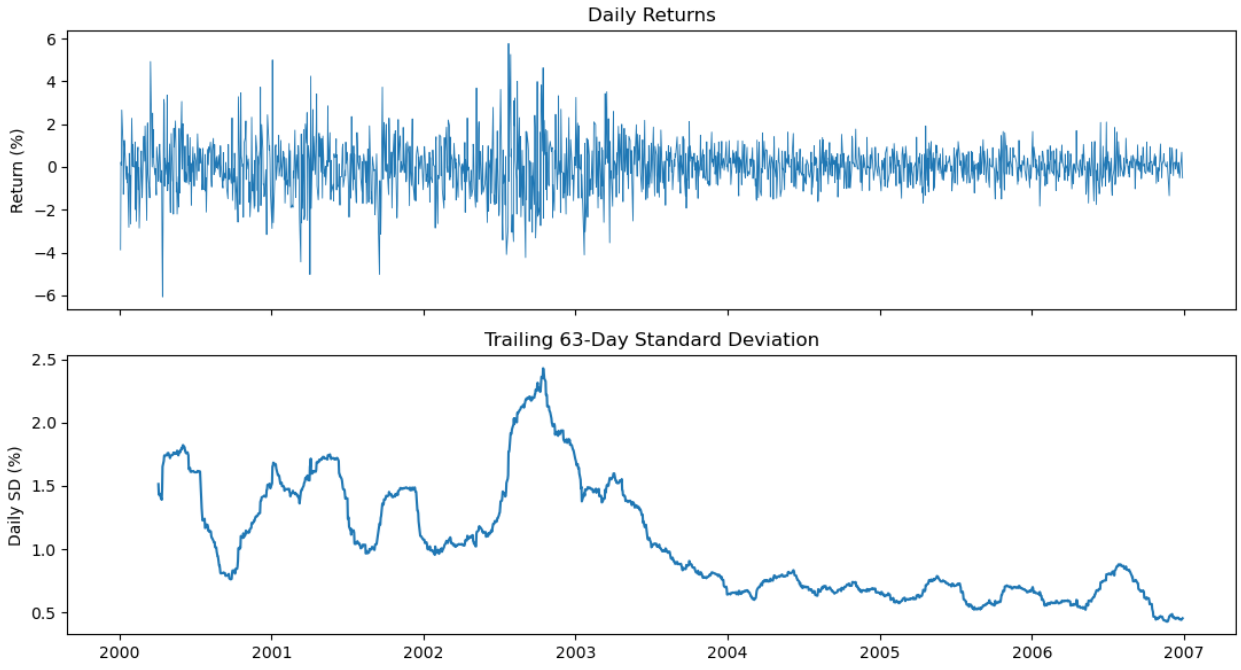


Figure 2: Daily returns and trailing standard deviation

The Ljung–Box test for returns through lag 10 gives a p-value of 0.1278, so the null of no autocorrelation through these lags is not rejected at the 5% level. Absolute returns, however, show positive autocorrelation across multiple lags, and their Ljung–Box test strongly rejects the null. The ARCH-LM test also strongly

rejects the absence of ARCH effects.

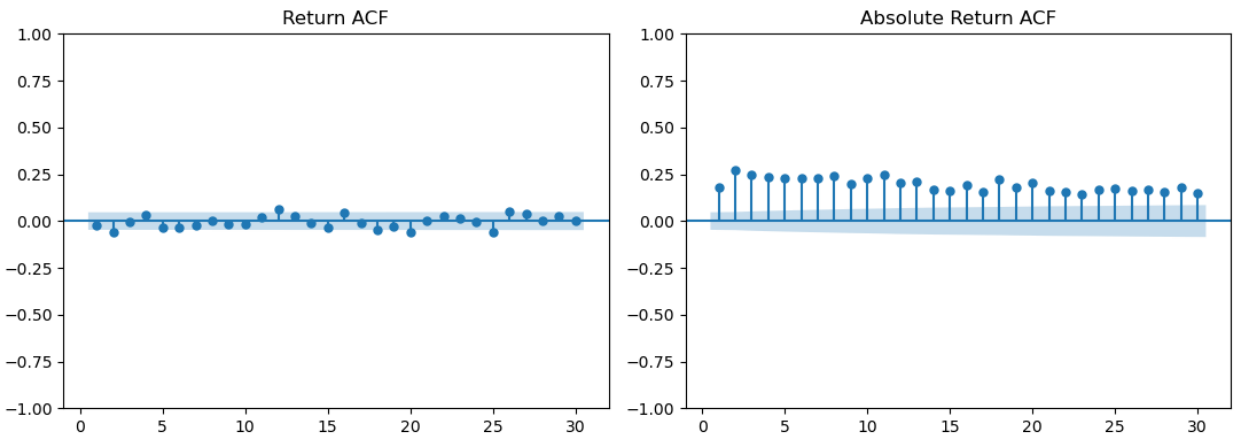


Figure 3: Return and absolute return ACF

Dependence in return magnitudes is therefore more pronounced than linear dependence in returns. This supports a dynamic variance model such as GARCH and the inclusion of historical volatility information in neural models.

1.4 Properties of the Realized Volatility Measures

Both log-transformed measures are highly persistent. Their first-order autocorrelations are close to 0.8 and remain above 0.5 around lag 30. This motivates considering current values and summaries over the previous week and month as candidate features.

However, the time series also exhibit changes in their levels across periods. High autocorrelation alone does not establish long memory or determine the optimal input length. Specific features should be selected through validation within the development period.

1.5 Association with Next-day Risk

Spearman rank correlations are calculated between current features and the next trading day's absolute return to examine potential predictive information. The correlation is 0.0982 for the current absolute return, compared with 0.3733 for log(rv5) and 0.3875 for log(bv). Both realized measures have stronger positive associations with next-day return magnitudes, supporting their consideration as predictors of future risk.

1.6 Implications for Model Selection

The findings motivate four candidate models. Historical simulation provides a simple benchmark for assessing whether additional complexity improves forecasts. GARCH(1,1)-t is motivated by volatility clustering, ARCH effects, and heavy-tailed returns.

The neural approaches are a quantile MLP and a quantile LSTM. Both use quantile loss to predict the three target quantiles directly. The MLP tests possible nonlinear relationships between the inputs and tail risk. LSTM is a type of recurrent neural network designed to process sequential data. The two neural models should use the same historical information wherever practical to assess the role of network architecture. All models will be evaluated within a common rolling forecasting framework.

2 Model Description

2.1 Non-Neural Approaches

2.1.1 Historical Simulation

Historical simulation estimates next-day VaR using the empirical distribution of recent returns.

At forecast date t, consider the latest W daily returns:

$$\mathcal{R}_t = (r_{t-W+1}, \ldots, r_t)$$

Sort these returns in ascending order:

$$r_{(1),t} \leq r_{(2),t} \leq \cdots \leq r_{(W),t}.$$

The next-day VaR forecast is the empirical lower-tail quantile:

$$\widehat{q}_{\alpha,t+1} = r_{(\lceil W\alpha \rceil),t}, \quad \alpha \in \{0.01, 0.05, 0.10\}.$$

The ceiling function rounds the rank upward to the nearest integer. For example, with 1,000 observations, the 1% VaR forecast is the 10th smallest return:

$$\widehat{\mathrm{VaR}}_{0.01,t+1} = r_{(10),t}.$$

The window moves forward one trading day at a time, adding the latest observed return and removing the oldest observation. The method assumes that the recent return distribution provides a useful approximation to the next day's distribution.

Remark 2.1. It does not directly use rv5 or bv, or separately forecast volatility.

2.1.2 GARCH

GARCH(1,1)-t forecasts next-day volatility and combines it with a Student-t distribution to obtain VaR. Returns are decomposed into a constant mean and a random shock:

$$r_t = \mu + \epsilon_t, \qquad \epsilon_t = \sigma_t z_t.$$

Here, μ is the conditional mean, ϵ_t is the return shock, σ_t is the conditional volatility, and z_t is a standardized innovation.

The innovations follow a Student-t distribution standardized to have zero mean and unit variance:

$$z_t = \sqrt{\frac{\nu-2}{\nu}}\, u_t, \qquad u_t \sim t_\nu, \qquad \nu > 2.$$

The degrees of freedom ν control tail thickness: smaller values imply heavier tails. The scaling factor adjusts the variance of the ordinary Student-t variable to one.The next-day conditional variance is given by:

$$\sigma_{t+1}^2 = \omega + a\epsilon_t^2 + b\sigma_t^2.$$

The parameter ω is the variance intercept. The term $a\epsilon_t^2$ captures the effect of the current shock, while $b\sigma_t^2$ captures the persistence of the current volatility level. The two lag orders in GARCH(1,1) correspond to one lag of the squared shock and one lag of the conditional variance.

After estimating the parameters within each rolling training window, the next-day VaR forecast is:

$$\widehat{\mathrm{VaR}}_{\alpha,t+1} = \widehat{\mu} + \widehat{\sigma}_{t+1}\sqrt{\frac{\widehat{\nu}-2}{\widehat{\nu}}}\, T_{\widehat{\nu}}^{-1}(\alpha), \qquad \alpha \in \{0.01, 0.05, 0.10\}.$$

Here, $T_{\widehat{\nu}}^{-1}(\alpha)$ is the quantile of the ordinary Student-t distribution. The forecast combines the estimated mean, predicted volatility, and the standardized lower-tail quantile. At the three target levels, this quantile is negative. Higher predicted volatility therefore produces a more negative VaR forecast, holding other quantities fixed.

Remark 2.2. The usual restrictions ensure positive variance and a finite long-run variance:

$$\omega > 0, \qquad a, b \geq 0, \qquad a + b < 1.$$

2.2 Neural Network Approaches

Both networks use daily returns and two log-transformed realized volatility measures:

$$\mathbf{x}_t = \begin{bmatrix} r_t \\ \log(rv5_t) \\ \log(bv_t) \end{bmatrix}.$$

The latest L trading days are used to predict three next-day return quantiles:

$$\widehat{\mathbf{q}}_{t+1} = \begin{bmatrix} \widehat{q}_{0.01,t+1} \\ \widehat{q}_{0.05,t+1} \\ \widehat{q}_{0.10,t+1} \end{bmatrix}.$$

Inputs are standardized using only the current training window.

Both networks use quantile loss. For a forecast error $u_{\alpha,i} = y_i - \widehat{q}_{\alpha,i}$, we define

$$\rho_\alpha(u) = \begin{cases} \alpha u, & u \geq 0, \\ (\alpha - 1)u, & u < 0. \end{cases}$$

The objective averages this loss over N training examples and the three target quantiles:

$$\mathcal{L} = \frac{1}{3N} \sum_{i=1}^{N} \sum_{\alpha \in \{0.01, 0.05, 0.10\}} \rho_\alpha\big(y_i - \widehat{q}_{\alpha,i}\big).$$

Each target y_i is the next trading day's return for the corresponding input. Both models are fitted within rolling training windows and produce one-day-ahead forecasts.

2.2.1 MLP

The MLP flattens the historical inputs in chronological order into a feature vector. A single-hidden-layer representation is:

$$\begin{aligned} \mathbf{v}_t &= \mathrm{vec}(\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t), \\ \mathbf{h}_t &= \phi(\mathbf{W}_1 \mathbf{v}_t + \mathbf{b}_1), \\ \widehat{\mathbf{q}}_{t+1} &= \mathbf{W}_2 \mathbf{h}_t + \mathbf{b}_2. \end{aligned}$$

The weights and biases are learned from the training data, and ϕ is a nonlinear activation function. The MLP examines possible nonlinear relationships between historical inputs and future tail risk.

2.2.2 LSTM

The LSTM processes the same historical inputs as the MLP in chronological order. At each step, it updates a hidden state and a cell state through three gates. For $s = t - L + 1, \ldots, t$, the gate values and candidate memory are:

$$\begin{aligned} \mathbf{f}_s &= \mathrm{sigmoid}(\mathbf{W}_f \mathbf{x}_s + \mathbf{U}_f \mathbf{h}_{s-1} + \mathbf{b}_f), \\ \mathbf{i}_s &= \mathrm{sigmoid}(\mathbf{W}_i \mathbf{x}_s + \mathbf{U}_i \mathbf{h}_{s-1} + \mathbf{b}_i), \\ \mathbf{o}_s &= \mathrm{sigmoid}(\mathbf{W}_o \mathbf{x}_s + \mathbf{U}_o \mathbf{h}_{s-1} + \mathbf{b}_o), \\ \widetilde{\mathbf{c}}_s &= \tanh(\mathbf{W}_c \mathbf{x}_s + \mathbf{U}_c \mathbf{h}_{s-1} + \mathbf{b}_c). \end{aligned}$$

The forget gate $\mathbf{f}_s$ controls how much previous memory is retained. The input gate $\mathbf{i}_s$ controls how much candidate information is added. The output gate $\mathbf{o}_s$ controls which parts of the transformed cell state enter the hidden state. Each sigmoid gate component takes a value between zero and one.

The cell state and hidden state are updated as follows:

$$\begin{aligned} \mathbf{c}_s &= \mathbf{f}_s \odot \mathbf{c}_{s-1} + \mathbf{i}_s \odot \widetilde{\mathbf{c}}_s, \\ \mathbf{h}_s &= \mathbf{o}_s \odot \tanh(\mathbf{c}_s). \end{aligned}$$

Here, $\odot$ denotes elementwise multiplication. The cell state carries memory across the input sequence, while the hidden state passes information to the next step and the final prediction. The final hidden state is mapped to the three next-day quantile forecasts:

$$\widehat{\mathbf{q}}_{t+1} = \mathbf{W}_q \mathbf{h}_t + \mathbf{b}_q.$$

The network is trained using the same quantile loss as the MLP.

In this project, positive autocorrelation in absolute returns and persistence in the realized volatility measures suggest that historical risk conditions contain useful information. The gating mechanism allows the LSTM to learn which historical information to retain and which new information to incorporate, potentially capturing both persistence and changes in volatility.

3 Rolling Window Settings and Computational Details

All four models use a fixed-length rolling window for one-day-ahead forecasting. After the observations for trading day t become available, the latest 1,260 trading days are used to estimate the model and forecast the next day's 1%, 5%, and 10% VaR. The training window is:

$$\mathcal{W}_t = \{t - W + 1, \ldots, t\}, \qquad W = 1260.$$

After the next observation becomes available, the window adds the latest observation and removes the oldest one. All models use the same forecast dates and historical window length. Parameters are re-estimated at each forecast origin, and neural networks do not inherit fitted weights from the previous window.

The validation period runs from January 25, 2005 to December 29, 2006, containing 488 forecast dates. The initial training window covers January 4, 2000 to January 24, 2005. The final test period runs from January 3, 2007 to June 27, 2018, containing 2,892 forecast dates. Model parameters continue to be re-estimated daily under the predefined rolling procedure, using only information available at each forecast origin.

The MLP and LSTM use the same three inputs: returns, log(rv5), and log(bv). Each training example uses 20 consecutive trading days, with the following day's observed return as its target. Each training window therefore contains $N_{\text{train}} = W - L = 1260 - 20 = 1240$ training examples. Standardization parameters are calculated only from the training input dates within the current window. After fitting, the latest 20 observations are used to forecastthe next day.

The current baseline configuration uses one ReLU hidden layer of width 16 for the MLP and one LSTM layer with a hidden-state dimension of 16. Both networks use Adam with a learning rate of 0.001, a batch size of 128, and 50 training epochs per window, without early stopping. Training examples are shuffled across batches. Both networks minimize the average quantile loss across the three target levels.

4 Results

4.1 Evaluation Metrics

Let N denote the number of evaluation dates, r_t the observed daily log return, and $\widehat{q}_{\alpha,t}$ the return quantile forecast made using information available through the previous trading day, where $\alpha \in \{0.01, 0.05, 0.10\}$. VaR is expressed as a lower-tail return quantile.

A VaR exception occurs when the observed return falls below the forecast:

$$I_{\alpha,t} = \begin{cases} 1, & r_t < \widehat{q}_{\alpha,t}, \\ 0, & r_t \geq \widehat{q}_{\alpha,t}. \end{cases}$$

All models are evaluated on the same dates and use consistent return units. The **exception count** and **failure rate** are defined as

$$K_\alpha = \sum_{t=1}^{N} I_{\alpha,t}, \qquad \widehat{p}_\alpha = \frac{K_\alpha}{N}.$$

Under correct coverage, the expected exception count is $N\alpha$, and the failure rate should be close to α. A rate above the target suggests risk underestimation, whereas a rate below the target suggests conservative forecasts.

The **violation ratio** is

$$\mathrm{VR}_\alpha = \frac{\widehat{p}_\alpha}{\alpha} = \frac{K_\alpha}{N\alpha}.$$

A ratio close to 1 indicates that the overall exception frequency is close to its target. However, frequency alone does not measure the severity or temporal clustering of exceptions.

Quantile loss evaluates the overall performance of the forecasts. Define the forecast residual as

$$u_{\alpha,t} = r_t - \widehat{q}_{\alpha,t}.$$

The daily quantile loss is

$$\rho_\alpha(u) = \begin{cases} \alpha u, & u \geq 0, \\ (1-\alpha)|u|, & u < 0. \end{cases}$$

The average quantile loss over the evaluation period is

$$\mathrm{QL}_\alpha = \frac{1}{N} \sum_{t=1}^{N} \rho_\alpha \left(r_t - \widehat{q}_{\alpha,t} \right).$$

For the lower-tail quantiles considered here, returns below the forecast receive a larger penalty, while excessively conservative forecasts also incur a loss. QL therefore accounts for both the direction and magnitude of forecast errors, rather than only counting exceptions.

Three statistical tests are conducted at the 5% significance level. The **unconditional coverage (UC) test** examines the null hypothesis that the exception probability equals the target level α. The **independence (IND) test** examines whether the probability of an exception depends on whether an exception occurred on the previous trading day. The **conditional coverage (CC) test** jointly assesses correct coverage and this independence condition.

For each test, the null hypothesis is rejected when the p-value is below 0.05. Non-rejection indicates insufficient evidence against the null hypothesis in the evaluation sample.

4.2 Results on Validation Set

The validation period runs from 25 January 2005 to 29 December 2006 and contains 488 forecast dates. All four models generate one-day-ahead VaR forecasts using rolling windows of 1,260 trading days and only information available before the forecast date.

The initial MLP forecasts exhibit quantile crossing on 74 dates, representing 15.16% of the validation dates. No crossings are observed for the other models. To ensure ordered quantiles, the three daily neural network forecasts are sorted in ascending order:

$$\widehat{q}_{0.01,t} \leq \widehat{q}_{0.05,t} \leq \widehat{q}_{0.10,t}.$$

This post-processing uses only the predicted values and does not use the realized return. The results below are based on these ordered forecasts. The LSTM validation forecasts were already ordered and therefore remain unchanged.

Table 1: Validation performance summary for four models at different levels

α	Model	Exceptions	Expected	Failure rate (%)	VR	QL
1%	HS	0	4.88	0.00	0.000	0.000309
1%	GARCH-t	3	4.88	0.61	0.615	0.000184
1%	MLP	5	4.88	1.02	1.025	0.000280
1%	LSTM	3	4.88	0.61	0.615	**0.000176**
5%	HS	1	24.40	0.20	0.041	0.000934
5%	GARCH-t	20	24.40	4.10	0.820	0.000697
5%	MLP	23	24.40	4.71	0.943	0.000714
5%	LSTM	18	24.40	3.69	0.738	**0.000686**
10%	HS	9	48.80	1.84	0.184	0.001469
10%	GARCH-t	43	48.80	8.81	0.881	**0.001139**
10%	MLP	42	48.80	8.61	0.861	0.001220
10%	LSTM	32	48.80	6.56	0.656	0.001165

HS produces failure rates far below the target levels. It also has the highest QL at all three levels, showing that fewer exceptions do not necessarily imply lower forecast loss.

LSTM achieves the lowest QL at the 1% and 5% levels, while GARCH-t achieves the lowest QL at the 10% level. LSTM has lower failure rates than GARCH-t at the 5% and 10% levels, indicating more conservative forecasts.

After quantile ordering, MLP has the failure rates closest to the targets at the 1% and 5% levels. However, its QL remains higher than that of GARCH-t and LSTM. This illustrates that matching the overall exception frequency and minimizing forecast loss are complementary objectives.

Remark 4.1. The validation period contains only 488 forecast dates, with an expected count of 4.88 exceptions at the 1% level. A small number of exceptions can therefore substantially change the estimated failure rate at this level. So we need to look into the test period for a broader understanding.

The validation stage is used to check the forecasting pipeline, compare preliminary model performance, and determine the quantile-crossing treatment. It identifies crossings in the MLP forecasts and motivates a sorting procedure that does not use realized returns.

No systematic hyperparameter search is conducted. All four models are retained for testing, with hyperpa-

rameters and post-processing rules fixed before the test period. Model parameters continue to be re-estimated according to the rolling scheme, but the settings are not adjusted using test results.

4.3 Results on Test Set

The test period runs from 3 January 2007 to 27 June 2018 and contains 2,892 forecast dates. All four models generate one-day-ahead VaR forecasts using the predefined rolling window of 1,260 trading days. Hyperparameters and quantile-ordering rules are fixed before testing, while model parameters are re-estimated as the window moves forward. Each forecast uses only information available before the forecast date.

Table 2: Test performance summary for four models at different levels

α	Model	Exceptions	Expected	Failure rate (%)	VR	QL
1%	HS	47	28.92	1.63	1.625	0.000586
1%	GARCH-t	50	28.92	1.73	1.729	**0.000366**
1%	MLP	88	28.92	3.04	3.043	0.000499
1%	LSTM	72	28.92	2.49	2.490	0.000463
5%	HS	165	144.60	5.71	1.141	0.001654
5%	GARCH-t	193	144.60	6.67	1.335	**0.001295**
5%	MLP	205	144.60	7.09	1.418	0.001356
5%	LSTM	173	144.60	5.98	1.196	0.001297
10%	HS	285	289.20	9.85	0.985	0.002421
10%	GARCH-t	337	289.20	11.65	1.165	**0.002067**
10%	MLP	346	289.20	11.96	1.196	0.002125
10%	LSTM	307	289.20	10.62	1.062	0.002074

GARCH-t achieves the lowest QL at all three quantile levels, followed by LSTM. However, their losses are very close at the 5% and 10% levels. At the 1% level, all four models have failure rates above the target. MLP and LSTM produce failure rates of 3.04% and 2.49%, respectively. Although GARCH-t achieves the lowest QL, its failure rate is still 1.73%. HS has the failure rates closest to the targets at the 5% and 10% levels, but its QL is the highest at all three levels.

The following table shows the result for the statistical test mentioned above. Bold entries indicate rejection of the null hypothesis at the 5% significance level.

Table 3: Statistical test for all four models on test set at three different levels

α	Model	UC	IND	CC
1%	HS	**0.0019**	**< 0.0001**	**< 0.0001**
1%	GARCH-t	**0.0004**	0.2864	**0.0010**
1%	MLP	**< 0.0001**	**0.0060**	**< 0.0001**
1%	LSTM	**< 0.0001**	**0.0418**	**< 0.0001**
5%	HS	0.0884	**0.0031**	**0.0030**
5%	GARCH-t	**< 0.0001**	0.3725	**0.0003**
5%	MLP	**< 0.0001**	0.4966	**< 0.0001**
5%	LSTM	**0.0186**	0.8320	0.0613
10%	HS	0.7942	**0.0004**	**0.0020**
10%	GARCH-t	**0.0038**	0.0565	**0.0025**
10%	MLP	**0.0006**	0.9172	**0.0028**
10%	LSTM	0.2742	0.6065	0.4816

At the 5% significance level, UC and CC reject the null hypotheses for all four models at the 1% VaR level.

Together with failure rates above 1%, these results indicate that all models underestimate extreme downside risk. Only GARCH-t is not rejected by IND at this level; the other models also exhibit dependence between exceptions on adjacent trading days.

At the 5% and 10% VaR levels, for HS, UC does not reject the null hypothesis, whereas IND and CC reject at both levels. For GARCH-t and MLP, IND does not reject at these two levels, but UC and CC reject. LSTM is the only model for which none of the three tests rejects at the 10% level, suggesting relatively good calibration at this level. However, at the 5% level, UC still rejects LSTM even though CC does not.

Also, for further reference, we plot the figures of the actual returns, VaR, and exceptions for all four models at different α levels.

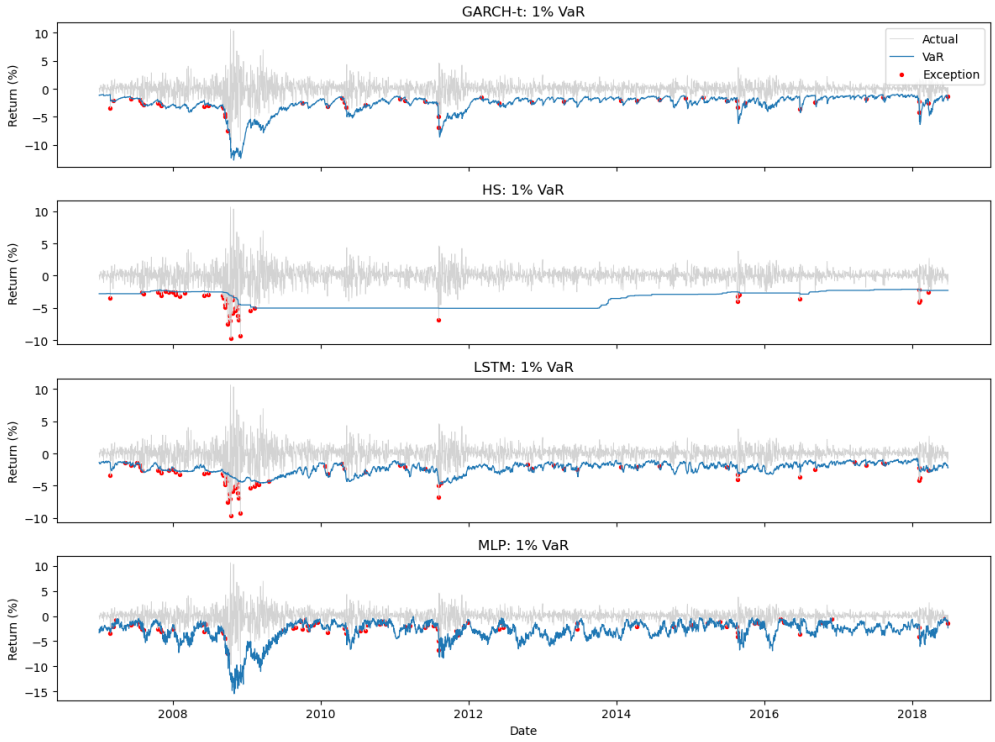


Figure 4: 1% VaR Forecasts and Exceptions Across Four Models During the Test Period

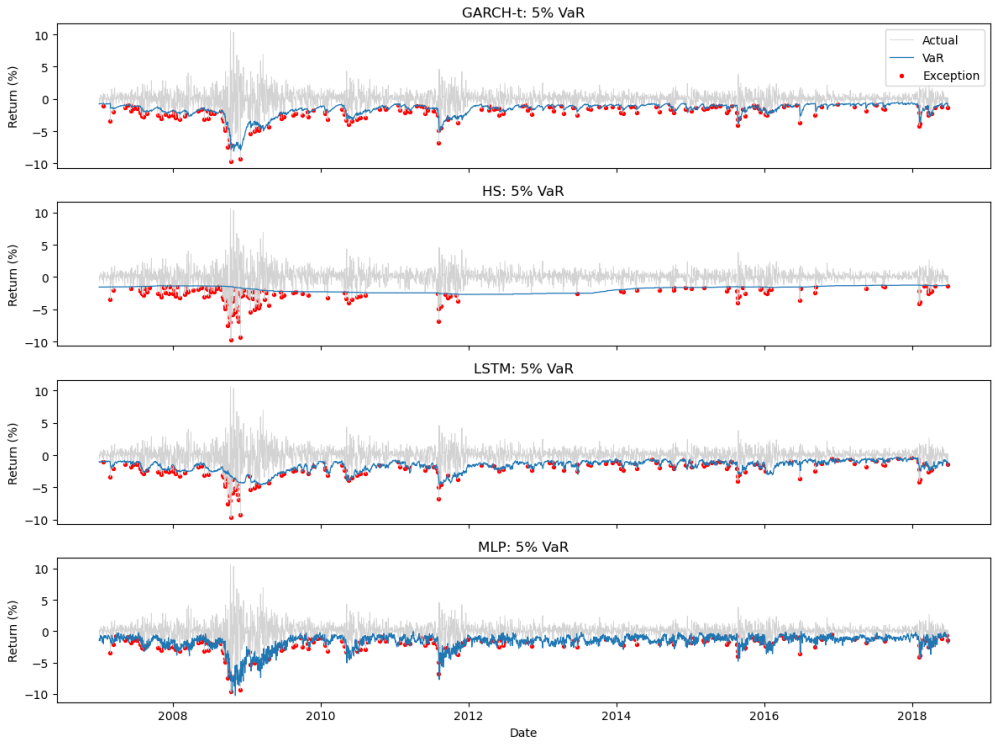


Figure 5: 5% VaR Forecasts and Exceptions Across Four Models During the Test Period

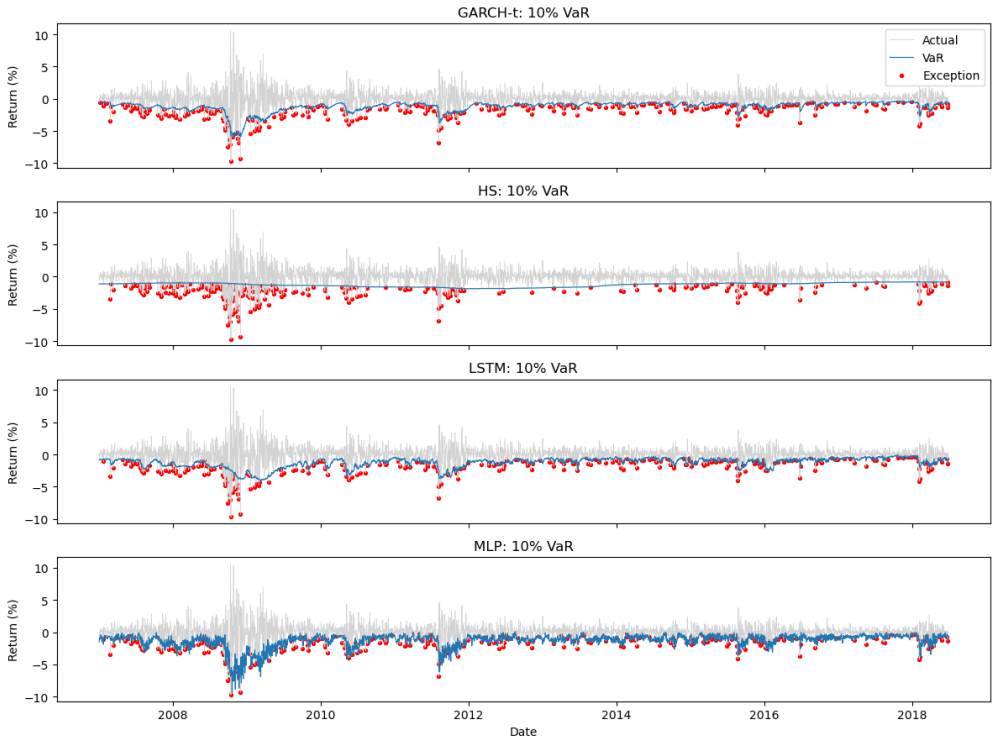


Figure 6: 10% VaR Forecasts and Exceptions Across Four Models During the Test Period

5 Comparative Discussion

5.1 Forecast Loss

No model dominates across all evaluation criteria. GARCH-t achieves the lowest test QL at all three quantile levels, but UC rejects at each level. Conversely, HS has failure rates closer to the targets at the 5% and 10% levels, but has the highest QL and is rejected by IND at both levels.

Model evaluation therefore requires joint consideration of forecast loss, overall exception frequency, and temporal dependence in exceptions. LSTM offers a relatively good balance at the 10% level, with near-minimum QL and no rejection by the three tests.

5.2 Model Structure and Forecast Performance

The forecast plots show that HS adjusts slowly and maintains unchanged thresholds for extended periods. This is consistent with its use of empirical quantiles from a long rolling window.

GARCH-t explicitly models time-varying volatility, allowing its forecast thresholds to adjust to changing risk conditions. This structure is consistent with the volatility persistence observed in the exploratory analysis and with its relatively low test QL.

LSTM has lower test QL than MLP at all three levels, suggesting that sequential processing of the historical inputs produces lower average loss under the present configuration. However, this does not establish general advantage of LSTM over MLP. Also, both neural models have failure rates substantially above the 1% target.

5.3 Scope and Limitations

LSTM's validation QL advantage at the 1% and 5% levels does not persist in the test period, indicating that model rankings can vary across evaluation periods. The test period covers a broader range of risk conditions. However, without a subperiod analysis, the ranking changes cannot be attributed entirely to a particular market episode.

HS and GARCH-t use only historical returns, whereas the neural models also use rv5 and bv. The study therefore compares complete forecasting specifications and does not isolate the contributions of network architecture or additional predictors.

Also, hyperparameters are not systematically tuned, and the conclusions are therefore specific to the current data and settings.

6 Conclusion and Potential Improvement

6.1 Conclusion

This study compares HS, GARCH(1,1)-t, MLP, and LSTM for forecasting next-day SPY return quantiles under a common rolling forecasting framework. GARCH-t achieves the lowest test QL at all three target

levels, while LSTM shows relatively good calibration at the 10% level.

However, all four models produce excessive exceptions at the 1% level, highlighting the difficulty of forecasting extreme downside risk. Overall, the more complex neural models do not consistently outperform the traditional models under the current settings.

6.2 Potential Improvement

First, neural network settings such as hidden-layer size, input sequence length, learning rate, and training epochs could be tuned using validation data. Training time should also be recorded to balance forecast performance and computational cost.

Second, neural networks using only historical returns could be compared with networks using returns, rv5, and bv, while keeping other settings as similar as possible. This would help assess the contribution of the additional volatility information. Comparison with traditional models would then provide a clearer understanding of the role of the neural network architecture itself.

Finally, the test period could be divided into different time intervals or high- and low-volatility periods. Comparing models across these periods would help identify whether their strengths vary with market conditions and inform model selection for different risk environments. Any new selection rule motivated by this analysis would require evaluation on new out-of-sample data.